$$\lim_{x \to \infty} \frac{2x^2 + 3x + 5}{3x^2 - 2x + 1}$$

$$\lim_{x \to \infty} \frac{\cancel{x^2}\left(2 + \frac{3}{x} + \frac{5}{x^2}\right)}{\cancel{x^2}\left(3 - \frac{2}{x} + \frac{1}{x^2}\right)}$$

$$\lim_{x \to \infty} \frac{2 + \cancel{\frac{3}{x}}^{0} + \cancel{\frac{5}{x^2}}^{0}}{3 - \cancel{\frac{2}{x}}^{0} + \cancel{\frac{1}{x^2}}^{0}} = \frac{2}{3}$$

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$$\lim_{x \to \infty} \frac{2x^2 + 7x + 5}{x^3 + 2x + 1}$$

$$\lim_{x \to \infty} \frac{\cancel{x^2}\left(2 + \frac{7}{x} + \frac{5}{x^2}\right)}{x^{\cancel{3}}\left(1 + \frac{2}{x^2} + \frac{1}{x^3}\right)}$$

$$\lim_{x \to \infty} \frac{2 + \cancel{\frac{7}{x}}^{0} + \cancel{\frac{5}{x^2}}^{0}}{x\left(1 + \cancel{\frac{2}{x^2}}_{0} + \cancel{\frac{1}{x^3}}_{0}\right)}$$

$$\lim_{x \to \infty} \frac{2}{x} = 0$$

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$$\lim_{x \to \infty} \frac{\sqrt{2x^2+1}}{x+3}$$

$$\lim_{x \to \infty} \frac{\sqrt{x^2\left(2+\frac{1}{x^2}\right)}}{x\left(1+\frac{3}{x}\right)}$$

$$\lim_{x \to \infty} \frac{\sqrt{x^2}\sqrt{2+\frac{1}{x^2}}}{x\left(1+\frac{3}{x}\right)}$$

$$\lim_{x \to \infty} \frac{\cancel{x}\sqrt{2+\cancel{\frac{1}{x^2}}^{\,0}}}{\cancel{x}\left(1+\cancel{\frac{3}{x}}^{\,0}\right)} = \sqrt{2}$$

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$$\lim_{x \to \infty} \frac{1}{x} \sin\left(\frac{\pi}{x}\right)$$

$$\left[\left(\lim_{x \to \infty} \frac{1}{x}\right)\left(\lim_{x \to \infty} \sin\left(\frac{\pi}{x}\right)\right)\right]$$

$$\frac{1}{\infty} \qquad \lim_{x \to \infty} \sin \pi \cdot \cancel{\frac{1}{x}}^{0} = 0$$

$$0$$

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$$\lim_{x \to -\infty} \frac{\sqrt{2x^2+1}}{x+3}$$

$$\lim_{x \to -\infty} \frac{\sqrt{x^2\left(2+\cancel{\frac{1}{x^2}}\right)^{0}}}{x\left(1+\cancel{\frac{3}{x}}\right)^{0}}$$

$$\lim_{x \to -\infty} \frac{|\cancel{x}|\sqrt{2}}{\cancel{x}(1)}$$

$$\lim_{x \to -\infty} \frac{-\sqrt{2}}{1} = -\sqrt{2}$$

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$$\lim_{x \to \infty} \left(\sqrt{x^2+2x} - x \right)$$

$$\lim_{x \to \infty} \sqrt{x^2+2x} - x \cdot \frac{\sqrt{x^2+2x} + x}{\sqrt{x^2+2x} + x}$$

$$\lim_{x \to \infty} \frac{\cancel{x^2} + 2x - \cancel{x^2}}{\sqrt{x^2+2x} + x}$$

$$\lim_{x \to \infty} \frac{2x}{\sqrt{x^2+2x} + x}$$

$$\lim_{x \to \infty} \frac{2x}{\sqrt{x^2\left(1+\frac{2}{x}\right)} + 1}$$

$$\lim_{x \to \infty} \frac{2x}{|x|\sqrt{1+\frac{2}{x}} + 1}$$

$$\lim_{x \to \infty} \frac{2}{\sqrt{1+\underbrace{\frac{2}{x}}_{0}} + 1} = \frac{2}{2} = 1$$

$\forall : x > \infty$

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$$\lim_{x \to -\infty} \left(\sqrt{x^2+2x} - x \right)$$

$$\lim_{x \to -\infty} \sqrt{x^2+2x} - x \cdot \frac{\sqrt{x^2+2x} + x}{\sqrt{x^2+2x} + x}$$

$$\lim_{x \to -\infty} \frac{\cancel{x^2} + 2x - \cancel{x^2}}{\sqrt{x^2+2x} + x}$$

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2+2x} + x}$$

$$\lim_{x \to -\infty} \frac{2x}{\sqrt{x^2\left(1+\frac{2}{x}\right)} + 1}$$

$$\lim_{x \to -\infty} \frac{2x}{|x|\sqrt{1+\frac{2}{x}} + 1}$$

$\forall : x < -\infty$

$$\lim_{x \to -\infty} \frac{2}{-\sqrt{1+\frac{2}{x}} + 1}$$

$$\lim_{x \to -\infty} \frac{2}{-2} = -1$$

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